



pgeo: generalisation and optimisation

Design, tuning, accuracy, capacity and security, measured against Pelias on open data
for the State of Maine

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Executive summary

The study report measured a geocoder for one state. This one covers making it a geocoder for any state, and what that exposed.

The work was prompted by a single instruction: build New York and see whether the documentation survives contact. It did not, and neither did the code. A build is now a named set of US states rather than a hardwired Maine, and getting there surfaced eleven defects, four of which would have shipped wrong answers rather than failing loudly.

Four regions have now been built: Maine, New York, Vermont plus New Hampshire, and Arizona plus Nevada. Each new one has found at least one general defect that the previous ones could not, which is the argument for building a fourth rather than asserting the third was enough.

The headline results:

	Before	After
Regions buildable	Maine	any US state, or several at once
Maine build time	1,152 s	309 s
New York build time	6,699 s	2,643 s (measured)
New York accuracy	88.4%	95.5%
Arizona + Nevada accuracy	84.8%	91.9%
Maine accuracy	95.8%	96.0%
Known-answer failures across four builds	4, unseen	0
Two builds of the same data	differed on 10,769 rows	identical
Maine features outside Maine	4,270, labelled “, ME, USA”	2,516, none foreign, none beyond 16 km
Commands to build a region by hand	25	1

Nothing in this work tunes anything per region. That was the constraint throughout, and it is the reason the accuracy gap between New York and Maine closed: every fix replaced a fact about Maine with a rule that reads the data.

1. Why a second region is a test and not a feature

A system with one instance of anything cannot distinguish a rule from a coincidence. Maine’s postal codes begin 039 to 049; Maine’s name is one word; Maine contains no town called Maine; Maine’s counties have names that no other state in the build shares, because there was no other state. Each of those was written into the code as though it were a property of geocoding.

Building New York cost about four hours of machine time and found:

Defect	Would it have failed loudly?
Labels hardcoded to , ME, USA	No. New York would have answered “Albany, ME, USA”
Region gid hardcoded to Maine’s	No. Every New York feature stamped with Maine’s id
County abbreviations matched by name only	No. New York’s Franklin and Washington counties given Maine’s abbreviations and FIPS
state_fips literal ‘23’ in /v1/address	No
Parser stripped only “me” and “maine”	No. “350 5th Ave, New York, NY” resolved to 350 New York Ave
Parser tested only the last <i>word</i> for a state	No. “101 Penbrooke Drive Penfield New York” answered 395 km away
A state name that is also a town	No. “Canoga, New York” answered “New York Yoga”
boundary.gid treated Maine’s region as “everywhere”	No
ZIP-prefix range 039-049	Yes, eventually - no New York postcode would validate
DuckDB sniffed quote=’ ’ on a 5.3 M-row file	Yes, at row 90,718
OpenStreetMap never clipped to the state	No. Maine held places in Nova Scotia

Nine of eleven produce a plausible-looking wrong answer. That is the argument for building the second region rather than reasoning about it.

2. What a build is now

A *build* is one or more US states, named in regions/regions.json. That file holds, per state, the bounding box, the Geofabrik extract, the Who’s on First region id, the GNIS file and the OpenAddresses sources,

for all fifty states and the District of Columbia. It is generated by `scripts/gen_regions.py` from the Census cartographic boundary file, the Who's on First distribution and the OpenAddresses source listing on GitHub - not typed from memory - and existing entries are never overwritten, so Maine's row still records what the measured build used.

Every step takes `--build: fetch_data.sh, pelias-prep, pgeo-load, pgeo_setup.sh, pgeo_rebuild.sh, pgeo_dump.sh, dev_web.sh`, and the Ansible `pgeo_build` variable. A build that is not the default gets its own database, containers and ports, so several run side by side on one workstation; three did while this was written.

2.1 Facts that became rules

Was	Is
ZIP prefixes 039-049 decide which ZCTAs are Maine's	The Census ZCTA-to-county file decides, by land area
'Maine', 'ME', ' ', ME, 'USA' written into every row	The feature's own county gives its state, through Who's on First's parent link
county_ref matched on county name	Matched on county name <i>and</i> state FIPS
Parser strips a trailing "me" or "maine"	Strips any state the build covers, by postal code or full name, in as many words as the name has
OpenAddresses and Who's on First fetched by the Pelias CLI	Fetched from their publishers; a pgeo build needs nothing from Pelias

The ZCTA change is worth dwelling on because the obvious fix was wrong. Replacing the prefix range with "is the Census internal point inside the state polygon" seemed cleaner and was tried first. It loses Peaks Island, Cliff Island, the Cranberry Isles, Bailey Island, Blue Hill and Boothbay Harbor, because the cartographic boundary file generalises the coast, and it gains 03579, which is in New Hampshire. Asking the Census which county holds most of each ZCTA's land is both simpler and right: Maine keeps its islands and gains 03579 legitimately, because 846 km² of it lies in Oxford County against 611 km² in Coos County, New Hampshire.

3. Performance: the build is 3.7 times faster

The build's cost was concentrated in one step. Every raw feature was matched against the administrative polygons with four point-in-polygon probes, one per placetype, on a single core - a county outline is tens of thousands of vertices, and each candidate cost a full `ST_Intersects`.

Measured on New York, 200,372 real points against 8,619 polygons:

Form	Time	Answers
Four probes against whole polygons	62.6 s	(the original)
One probe against whole polygons	39.9 s	identical
One probe against <code>ST_Subdivide</code> pieces	7.8 s	identical

The first idea - one probe instead of four - was worth 1.6 times. The win was cutting the polygons into pieces of at most 128 vertices, so a probe touches a fragment rather than an outline. `full_area` carries the original polygon's size, because "smallest containing polygon" has to mean the smallest whole one.

Effect on a whole build:

	Before	After
Maine, entire build	1,152 s	309 s
Maine, the indexing step alone	~18 min	~4 min

The change is neutral, and was checked rather than assumed: rebuilding Maine produced the same 906,102 rows, and of those, **zero** differed in neighbourhood, locality, localadmin, county, region, region abbreviation, label or hierarchy JSON.

4. Correctness found on the way

4.1 A geocoder for Maine held places in Canada

Geofabrik's state extracts reach past the border, and nothing clipped them. Maine held 4,270 OpenStreetMap features outside Maine, every one labelled , ME, USA because the region was a build-wide constant: 22 Chemin Martin, Sainte-Anne-de-Madawaska and Gagetown in New Brunswick, Yarmouth Ferry Terminal in Nova Scotia 118 km away.

OpenStreetMap was taken to be the only source that needed clipping: Who's on First comes from the region's own descendants, GNIS from the state's own file, Overture is clipped during preparation, and ZCTAs are assigned by land area, so only OpenStreetMap arrives as a rectangle.

That reasoning was checked afterwards and is wrong at the edges. A state's own GNIS file includes features on its border whose recorded point falls just outside the polygon, and OpenAddresses admits a similar margin. Measured across the four builds, between 1,689 and 3,322 features sit outside the region - and in a single-state build the fallback that fills in a missing county labels every one of them with the build's state, which is the defect this section describes, returning at a twentieth of its old size. New York holds Ringwood River, NY, USA for a river in New Jersey, and Canadian parts of the Akwesasne reserve. The numbers, the distinction between single- and multi-state builds, and the fix are in `TOD0.md` section 10.

The clip uses Who's on First's region polygon, buffered by about 300 m, and the choice of polygon matters for the same reason it mattered for ZCTAs. Verified, not assumed:

- all 276 features on Peaks Island, Cliff Island and the Cranberry Isles survive;
- all 22 French and Wabanaki place names checked survive - Madawaska, Frenchville, Fort Kent, Van Buren, Grand Isle, St. Agatha, St. Francis, Isle au Haut, Mattawamkeag, Passadumkeag, Meddybemps, Wytopotlock, Millinocket, Allagash, Chesuncook, Norridgewock, Skowhegan, Wiscasset, Ogunquit, Damariscotta, Sinclair, Presque Isle.

The clip is by geometry and never by name. The St. John valley is French on both sides of the border and Madawaska is a town in each; a name-based rule would have been a trap. Both known-answer sets now carry a border-country place so this stays checked.

4.2 Null island

With the foreign features gone, the furthest thing from Maine was 8,206 km away: eight Who's on First postcodes with coordinates of exactly (0, 0), off the coast of Africa, labelled , ME, USA. Null island is missing data, not a location. Both staging paths now drop it, and nothing in a Maine build now sits more than 15 km outside the state.

4.3 Two real addresses merged into one

The address deduplication kept one row per house number, street and town. Normalisation makes “18 N St” and “18 North St” the same key, and Bangor has both, 5 km apart - so the build was merging two real addresses and discarding one. The tie-break only decided which one was lost.

Rows sharing a key are now clustered by proximity at about 200 m before a winner is chosen. **Maine gains 4,088 addresses** that were being discarded as duplicates of something a mile away.

Within a cluster the majority point wins. OpenAddresses publishes one row per unit, so a block of flats arrives as several rows at one coordinate and occasionally one row somewhere else: 101 Hancock St, Rumford has eight rows, seven agreeing and one 25 m away, and the build had been keeping whichever the table happened to hold first.

4.4 Determinism

That “whichever the table happened to hold first” had a second consequence. Two builds of the same data disagreed on **10,769 rows**, so a rebuild could silently move an address with nothing to show why. With the source id as a final tie-break, two consecutive builds now produce 908,347 rows with **identical** gids.

5. Accuracy

Accuracy is measured against ground truth taken from the source data, not from another engine: OpenAddresses points, Who's on First town labels, GNIS features, Overture places, Census ZCTAs. Maine's set is 1,560 cases; each of the other three is 198, built the same way. Where all four stand at the end of this work:

Group	Maine	New York	VT + NH	AZ + NV
Overall	96.0%	95.5%	92.4%	91.9%
addresses	97.8%	92.9%	91.4%	98.6%
towns	96.4%	92.9%	85.7%	89.3%
venues	94.2%	96.4%	92.9%	82.1%
reverse	100.0%	100.0%	96.2%	92.3%
lakes and summits	86.7%	100.0%	89.5%	78.9%
ZIP codes	98.3%	100.0%	100.0%	87.5%
misses	94.7%	94.7%	100.0%	100.0%

Maine is highest because Maine is where the tuning was measured, and the spread between the four is now four points rather than eleven. No setting in the engine names a region.

5.1 New York: 88.4% to 95.5%, with no New York setting

Group	First run	After the parser fixes	After the full rebuild	Maine
Overall	88.4%	93.9%	94.9%	95.9%
addresses	87.1%	91.4%	92.9%	97.8%
towns	60.7%	89.3%	89.3%	95.6%
venues	96.4%	96.4%	96.4%	94.2%
reverse	96.2%	96.2%	100.0%	100.0%

Group	First run	After the parser fixes	After the full rebuild	Maine
lakes and summits	100.0%	100.0%	100.0%	86.7%
ZIP codes	100.0%	100.0%	100.0%	98.3%
misses	94.7%	94.7%	94.7%	94.7%

The last column of movement is the rebuild of 2026-09-23, which carried the duplicate-address clustering, the OpenStreetMap clip and the five-digit house-number fix together. It recovered 35,251 addresses that the old duplicate key had merged away - Maine's equivalent was 4,088 - and removed 434 venues and 223 streets that the clip found outside the state. New York finished at 7,246,923 features in 2,643 s.

Towns were the failure, and the cause was the parser rather than the ranking. "Canoga, New York" was read as a place inside New York City, found nothing there, fell back to matching names across a million venues and answered "New York Yoga". What precedes the state name now decides which reading is meant: "Canoga" is a town, so New York is the state; "350 5th Ave" is an address, so New York is the city and survives as the locality.

Addresses recovered when the parser learned that a state name can be more than one word. Eleven of the fifty states have multi-word names, and only the last word had been tested.

The residual gap to Maine is in addresses, and is a property of density rather than a bug: New York holds 6.0 M address points to Maine's 0.7 M, so there are more near-identical candidates for any query.

5.2 Maine was not disturbed

Every change in this document is general, so Maine is the control. Across the whole sequence:

Stage	Maine accuracy	Cases changed
Before any of this work	95.8%	-
Parser generalised	95.7%	0 of 1,560, case by case
Point-in-polygon rewritten	95.7%	2, one each way
OpenStreetMap clipped to the region	96.0%	3
Duplicate addresses clustered	95.9%	1
Five-digit house numbers fixed	95.9%	0 of 1,560, in either direction

The one case lost to the duplicate-address change is 665 Saco St, Westbrook, whose ground truth is one arbitrary row among roughly 250 that OpenAddresses tags with that house number across a kilometre of street. The build now answers with the point 20 of them agree on.

5.3 Arizona and Nevada: a bug two regions could not see

Arizona plus Nevada was the fourth region, chosen because it is unlike the first three: desert rather than forest, Spanish and Diné place names rather than French and Abenaki, and a grid of young cities rather than New England's towns. It scored **84.8%** on first measurement against 198 cases, and addresses were the outlier at 77.1%.

Every one of the sixteen failing address cases began with a five-digit house number, and every case that began with a five-digit house number failed. A correlation that clean names the cause before any fix confirms it. The parser was reading the house number as a postcode:

```
SELECT (geocode.parse_rule('13023 E LIMA ST, PRESCOTT VALLEY')).*;
name      | housenumber | street      | locality      | postcode
E LIMA ST |             |             | PRESCOTT VALLEY | 13023
```

With no house number the address lookup found nothing, so the search fell back to the street and answered East Lima Street **at confidence 1.0** while 13023 E LIMA ST sat in the build. That is the worst shape a geocoding error can take: not a miss, which a caller can handle, but a confident wrong answer a kilometre up the road.

The reason Maine and New York never showed it is regional convention. New England house numbers rarely reach five digits; Phoenix and Las Vegas number outward from a city-centre baseline and reach five digits routinely. The parser now looks for a postcode only past a leading house-number token, in both implementations - the assumption had been written twice.

Group	First run	After the fix	Maine
Overall	84.8%	92.4%	95.9%
addresses	77.1%	98.6%	97.8%
towns	89.3%	89.3%	95.6%
venues	85.7%	85.7%	94.2%
reverse	92.3%	92.3%	100.0%
lakes and summits	78.9%	78.9%	86.7%
ZIP codes	87.5%	87.5%	98.3%
misses	100.0%	100.0%	94.7%

Addresses now score higher than Maine's, and no other category moved by a single case. Maine re-scored 95.9% with zero changed outcomes across all 1,560 of its cases. A fix that moves one region by 7.6 points and three others not at all is a fix to a rule, not a tuning for a region.

The residual gap is in the categories the fix does not touch: lakes and summits at 78.9%, where the South-west's washes and buttes are more densely and less distinctly named than Maine's ponds, and towns at 89.3%. That second one turned out to be a ranking defect, and section 5.4 is what happened when the test was sharpened enough to see it.

5.4 The test that was passing four wrong answers

Known answers were compared as substrings, so "Albany" matched the city of Albany, Albany County, and a shop called Albany, NY - Albany.com equally well. Pinning the expected layer turned four passes into failures across three builds, and they were two distinct defects.

A misspelling lost the town entirely. Not outranked - absent:

Query	Answered with	Should be
Albny, NY	Albany, NY - Albany.com, a shop in Colonie	Albany the city
Manchestr, NH	Manchester, NH, a venue	Manchester the city
Tuscon, AZ	One Hope - Tuscon, in Valencia West	Tucson

The cause was in the parser, not the ranking. A town was recognised only by exact spelling, so a misspelling parsed to nothing at all and the raw string went to name matching - where venues win, because business

listings carry their own town in their name (“Albany, NY - Albany.com”) and so match the string “Albny, NY” better than the bare name “Albany” does. Maine never showed it because Maine has fewer such venues.

The parser now falls back to a typo search over the build’s towns: one edit, or two when the two strings have the same letters. That second case is a transposition, Tuscon for Tucson, and it is the one that most needs catching, because a transposition shares almost no trigrams with its own word - the search could not have found Tucson from “Tuscon” however it was ranked. The limits matter as much as the rule: whole remaining text only, five characters or more, and only after an exact match fails. Without them walmart becomes Balmat, New York, and central park becomes the town of Parks.

A county outranked the city of the same name. Albany, New York returned Albany County and york returned York County, decided by importance alone - 1.000 against the city’s 0.942, worth 0.003 of score - while both rows read Albany, NY, USA and the user could not tell them apart. County and locality now share a deduplication bucket, so the pair collapses to one result and the county loses.

	Before	After	Cases changed
Maine	95.9%	96.0%	2 improved, 0 regressed
New York	94.9%	95.5%	1 improved, 0 regressed
Arizona + Nevada	92.4%	91.9%	0 improved, 1 regressed
Known-answer failures, all four builds	4	0	-

The one Arizona regression is Mesaa, Arizona, a generated case whose ground truth is a venue named “Mesa” inside the city of Mesa, scored against a 300 m radius. The query now answers with the city, which is what someone typing it means, and the case counts that wrong because the city centre is further than 300 m from that venue. It is reported here rather than argued away: the case set is the measure, and being confident about one of its cases is not a reason to edit it.

6. Multi-state builds

A build may cover several states. Vermont plus New Hampshire was built through the documented command, unattended: 1,074,744 raw features from two OpenStreetMap extracts appended into one schema, 285,846 on the Vermont side and 121,554 on the New Hampshire side, and all sixteen known answers passing on both front ends - including “Burlington, Vermont” giving Burlington, VT, USA and “Concord, New Hampshire” giving Concord, NH, USA, which is the thing a multi-state build has to get right and a check on the town name alone would not notice.

It found one defect the single-state builds could not: GNIS publishes a feature on a state line in both states’ files with the same id, and Vermont and New Hampshire share 36 of them, all brooks along the Connecticut River. The export validator refused the duplicate ids, which is how it was found rather than shipped.

Arizona plus Nevada was the second multi-state build and a much larger one, run start to finish through the single documented command:

	Vermont + New Hampshire	Arizona + Nevada
Features	731,596	5,008,365
split	464,500 VT / 264,626 NH	3,710,624 AZ / 1,296,052 NV
features with no state	2,470	1,689
Database	797 MB	4.7 GB
Build time	215 s	1,787 s

	Vermont + New Hampshire	Arizona + Nevada
Known answers	16 of 16	10 of 10
Accuracy	92.4%	91.9%

The split is the check that matters. A multi-state build that quietly attributed everything to the first state would still answer most queries plausibly; 3.7 M against 1.3 M, in a build whose region boundaries were never told which state is which, is the evidence that it does not. The features carrying no state at all - 1,689 of five million - are the border margin described in section 10, not a misattribution between the two states.

Both builds now pass their known answers in full, on both front ends, under the layer-pinned check of section 5.4. Vermont plus New Hampshire was rebuilt on 2026-09-23 to pick up the OpenStreetMap clip and the null-island fix it predated; it now builds in 215 s rather than 993, which is the ST_Subdivide change of section 3 reaching a region that had not been rebuilt since.

That rebuild also produced the sharpest operational lesson of this document, and it was my own mistake. I started it twice by accident, and the two builds destroyed each other: each begins by dropping and recreating the staging schema, so the second pulled the ground out from under the first mid-load. One process died with a traceback, the other printed `build complete in 206s`, and the result was missing 40% of its streets - Vermont 193 against New Hampshire's 40,300, with no Main Street in Burlington. A clean single rebuild gives 67,901. Nothing is wrong with the loader; what is missing is a lock, and a build that refuses to call itself complete when a loader wrote errors it exited zero on (`TOD0.md` section 13).

Vermont plus New Hampshire's 16 of 16 deserves a caveat, because verifying this section found the check to be weaker than it looks. Known answers are matched as substrings, so when that build's API container was left on an older image and answered "Burlington, Vermont" with a venue named Burlington Vermont, the check passed: the word "Burlington" was in the label. Three of its four town queries were returning venues while the build reported full marks. The image is now current and the answers are genuinely localities, but the lesson belongs in the record - a substring test cannot tell a town from a shop named after it, and the two front ends hold two copies of the parser with nothing checking they agree (`TOD0.md` section 11).

7. Corner cases across the fifty states

Surveyed by asking the registry and Who's on First, rather than by building fifty times.

Corner case	States	Handling
Name of two or more words	11: DC, NC, ND, NH, NJ, NM, NY, RI, SC, SD, WV	The parser tries the last n words
Name is also a town inside the state	3: MO, NY, WY	What precedes it decides
Abbreviation is an ordinary word	8: HI, ID, IN, LA, ME, MT, OK, OR	Only a trailing token is treated as a state
Box crosses the antimeridian	1: AK	Refused , with the reason
County name repeats in another state	many	The state must match too
ZIP prefix shared with a neighbour	many	A prefix must belong to this build alone
Feature on a state line	any multi-state build	Duplicates collapsed on the feature id

Alaska is the one refusal. Its box runs -179.3 to 179.95, so as minimum and maximum longitude it spans the planet: the Overture prefilter would pull the world and the coordinate check would accept anything. Handling it means carrying two boxes through every step.

8. Resource requirements

Measured on the same workstation the study used.

	Maine	New York
Features	908,347	7,213,395
Downloads	0.6 GB	3.0 GB
Database	1.0 GB	7.1 GB
On disk including WAL	3.6 GB	11 GB
Deployable dump	117 MB	890 MB
Build time	5 min	~30 min

Who's on First adds a one-off 5.2 GB shared by every build. A server restores the dump rather than building, so it needs the database plus room for the restore: 10 GB for a state the size of Maine, 20 GB for one the size of New York.

The capacity figures in the study - 32 concurrent users on one vCPU and 1 GB - are Maine's. New York has not been load-tested. What changes with size is the working set: New York's indexes alone are 1.8 GB, so a 1 GB machine reads from disk on most queries. Single-query latency on the workstation is 69 ms for `buffalo`, 123 ms for `albany` and 476 ms for a full street address, against 15-40 ms for Maine's equivalents.

`docs/SIZING.md` carries the full table.

9. What this means for the study report

The study's measurements stand. They describe a Maine build of pgeo 0.10.0 against Pelias, Photon and Nominatim, and nothing here changes that comparison: the accuracy figure moved from 95.8% to 95.9% through work that was neutral by construction and verified case by case.

Three things the study said should now be read differently:

- **"Ranking stays tuned on Maine."** It is still tuned on Maine, but the evidence is now that the tuning transfers: New York reaches 93.9% with no New York-specific setting, and the remaining gap is in addresses, where density rather than ranking explains it.
- **Build time.** The study's build timings are for the pre-optimisation pipeline. A Maine build is now 309 s rather than 1,152 s.
- **Feature count.** Maine now holds 908,347 features rather than 906,101: one ZCTA the land-area rule adds, 4,088 addresses the duplicate clustering recovers, less what the clip removes.

10. What is not done

- **The border margin.** Section 4.1 clipped OpenStreetMap to the region polygon, but the other sources are still filtered by the build's bounding box alone, so a margin of out-of-region features survives: 1,689 in Arizona plus Nevada, 514 of them more than a kilometre out. GNIS is the worst at 297 features up to 66 km outside, in Utah, New Mexico and Sonora; OpenAddresses adds 240. They carry no state, so they label honestly rather than falsely - the Maine defect of section 4.1 was worse precisely because those features claimed to be in Maine - but they are still findable and should not be there. The same polygon clip already written for OpenStreetMap applies unchanged. A separate 1,037 OpenStreetMap streets are an artifact rather than a leak: a street that crosses the border is kept whole and its representative point is its centroid, which can fall outside.

- **The Who's on First ancestry gap.** Some places carry no region ancestor in Who's on First and are dropped: 1 in Maine, 5 in New York, 62 in Nevada, 72 in Arizona. The loss is not evenly distributed - it falls on unincorporated and tribal communities (Chutum Vaya, Allenville, Cibecue Creek, Jarbidge), which is the population least well served by every other geocoder too. A point-in-polygon fallback against the region geometry would recover them.
- **Venues outrank the localities they are named after.** "Tuscon, AZ" answers One Hope - Tuscon and "Albany, New York" answers Sims Metal - Albany. This is the only known-answer failure left in any build, and it is a ranking question: a near-exact name match on a venue currently beats an exact match on a locality.
- **Query-time tie-breaking** is still by physical row order, worth about one case of accuracy noise between builds.
- **Alaska** is refused rather than handled.
- **A bounding-box build** - part of a state, or the adjoining parts of several - would scale every cost down together. The plumbing is already threaded, since every step reads the build's box.
- **Container images** remain designed and unpublished.
- **New York has not been load-tested**, so its capacity is unknown.